DATA COMMUNICATION

Er. Prabhakar Singh
Everest Engineering college
Sanepa -2 ,Lalitpur

- **Text Book**
  - **William Stallings, Data and Computer Communications, 10th edition**
  - **Behrouz Forouzan , Introduction to Data Communications and Networking**

- **References**
  - **U.D. Black, Data Communications and Distributed Networks,**
  - **Oppenheim, Signals and Systems**

Chapter 1
Introduction

1.1 Evolution of Data Communication

1.2 Analog and Digital Data Transmission, Data Communication Terminology

1.3 Standards Organizations, Applications

Definition- Data Communications

- Data – “Datum” derived from Latin means “Something Given” in mid 18 Century.
- Communication- “Communicare” from Latin means “to share”.
- **Data communication** refers to the exchange of data between a source and a receiver.
- Data is the entity which conveys some meaning based on some manually agreed rules or communication.
- Data is the information what we had want to sent.
- For example if a sender sends 01000001 to receiver it has no meaning unless the receiver understand. But if it is said that an ASCII character ,the receiver Can under stand it is A.

- The meanings of source and receiver are very simple. The device that transmits the data is known as source and the device that receives the transmitted data is known as receiver.
- The Characteristics of Data Communication
  - Delivery
  - Accuracy
  - Timeliness

Data is always transfer from the transmitter to receiver in the form of signal. It is of Two types:

- Digital Data
- Analog Data

1)Analog Data: Information that has sent has continuous(infinite) values in some interval of time. Analog data use values that change very smoothly.

- A good example of this is an analog clock. An analog clock shows the time with a smoothly moving seconds hand. The change is continuous.
- Sound is also a good example of analog data. Sound waves change in a very smooth way.

All analogue devices use analog data. Examples of analog devices include:

Microphone

•Headphones Loud Speaker

•Sensors (temperature,pressure

etc)

2)Digital Data: Information that has finite number of value within a certain time is digital data.

- Digital data jumps from one value to the next in a step by step sequence.
- A good example of this is a digital clock.
- A digital clock jumps from one second to another in clear steps. The change is not smooth or continuous.
- Information stored in memory of computer.
- All digital devices use digital data.

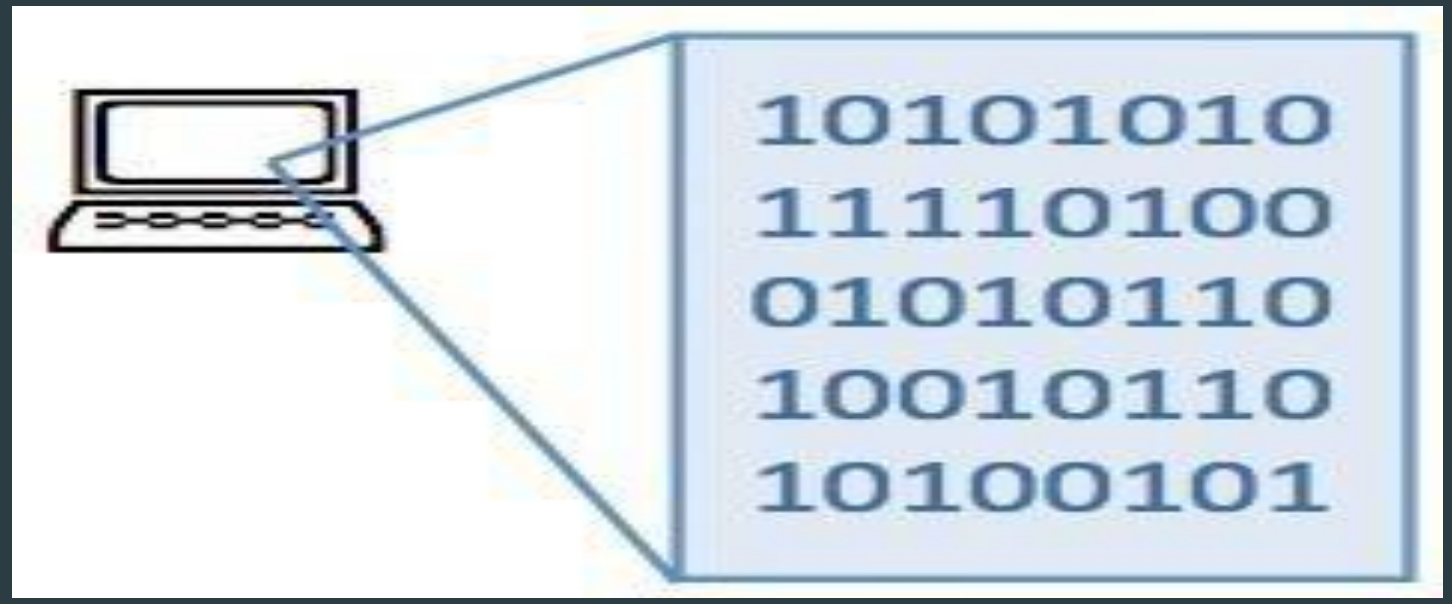


Fig: Storing the digital data

Signal:

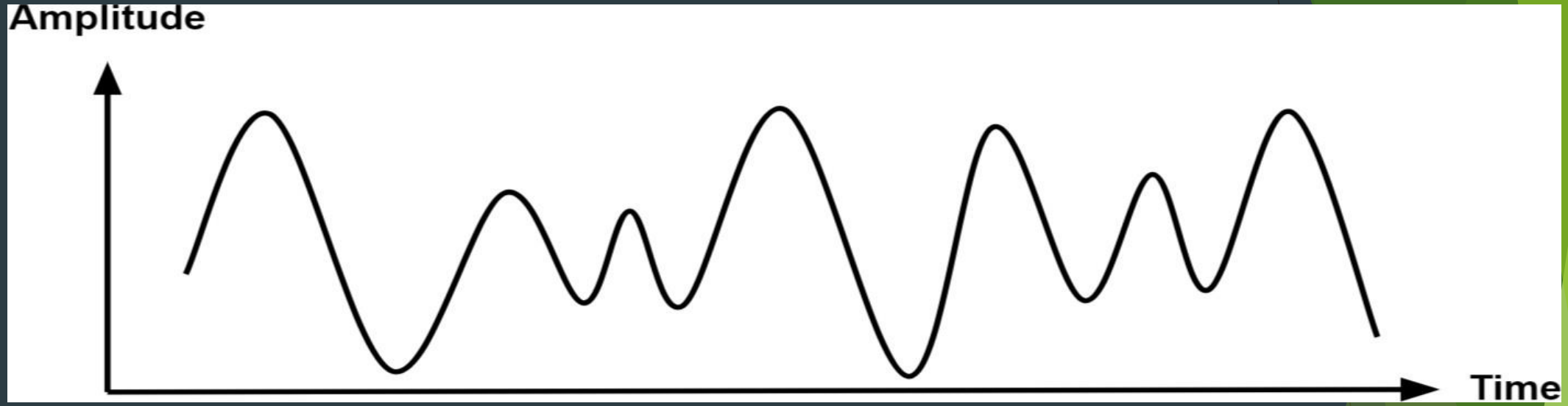
- A signal is a function that represent the variation of physical quantity with respect to any other parameter.
- A signal is the functional representation of a physical quantity or variable, and it contains information about the behavior of the physical quantity.
- A signal is represented as a function of an independent variable t. Most of the time t stand for time. The mathematical notation of a signal is x(t).
- When ever data has to be sent over communication medium it has to be converted into signal .
- Signal is an electric , electronic and optical representation of data which is sent over communication channel.
- There are the two types of signal

a) Analog Signal

b) Digital Signal

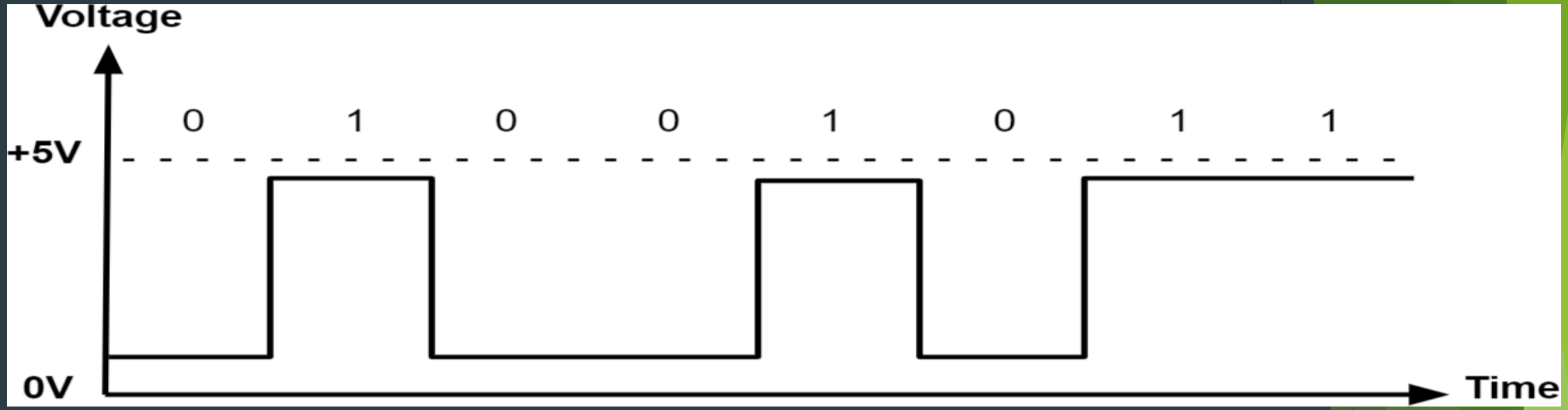
1. **Analog Signal:**

- Analog signal is a continuous signal in which one time-varying quantity represents another time- based variable.
- These type of electronic signals are time-varying
- Minimum and maximum values which is either positive or negative.
- Analog Signal works on continuous data.
- When plotted on a voltage vs time graph, an analog signal should produce a smooth and continuous curve. There should not be any discrete value changes.
- Electrical Signal Obtained from microphone ,photodetector cell example of analog signal


Amplitude
Time

2. **Digital Signal:**

- A digital signal is a signal that represents data as a sequence of discrete values.
- A digital signal can only take on one value from a finite set of possible values at a given time.
- With digital signals, the physical quantity representing the information can be many things.
- Digital signals are used in all digital electronics, including computing equipment and data transmission devices.
- When plotted on a voltage Vs time graph, digital signals are one of two values, and are usually between 0V and VCC


Voltage
0
1
0
0
1
0
1
1
+5V
0V
Time

Data Representation:

Information today comes in different form such as Text ,Number, Images, audio & video.

a)**Text**: In data communication text is represented as bit, or sequence of bit.

Different set of bit pattern have been designed to represent text symbols Each set of code and various coding technique are Unicode, ASCII etc.

b)**Numbers**: Number are also represented by bit pattern and used to simplify the mathematical operation.

c)I**mages**: An image is composed of matrix of pixels ,Where each pixel is small dot and is assigned a bit pattern.

d)**Audio** : It is refers to the recording of sound or image.

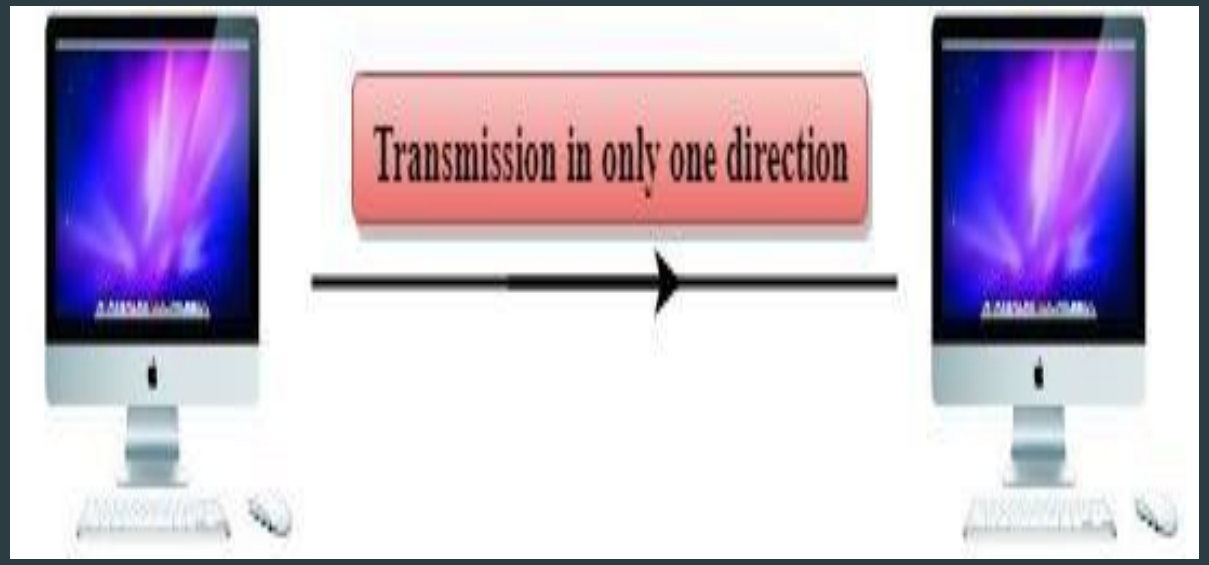
e)**Video**: It refers to the broadcasting of pictures or movies. Video can either be produced as continuous entity or the combination of images ,each of entity arranged to convey the idea of motion.

Data Flow

- The way in which data is transmitted from one device to another device is known as transmission mode.
- Transmission mode between two device can be either simplex, half-duplex and full duplex depending upon Flow the data.

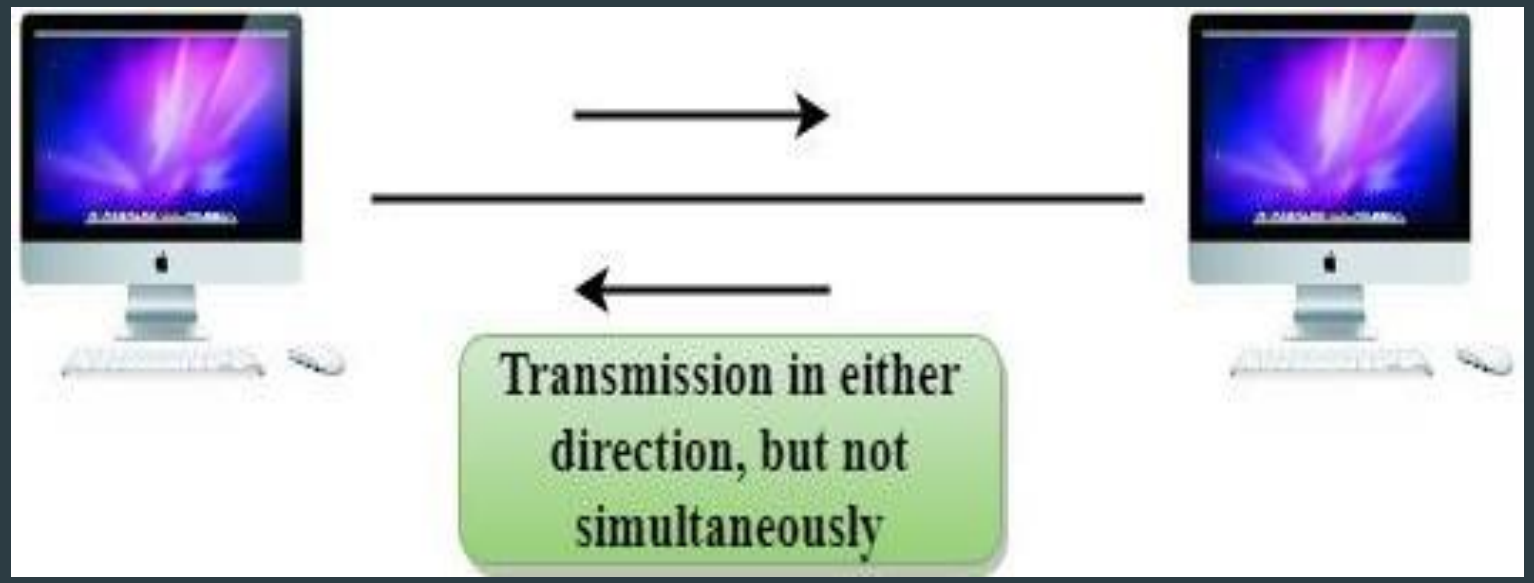
1. **Simplex mode:**

- In Simplex mode, the communication is unidirectional, i.e., the data flow in one direction.
- A device can only send the data but cannot receive it or it can receive the data but cannot send the data.
- This transmission mode is not very popular as mainly communications require the two-way exchange of data. The simplex mode is used in the business field as in sales that do not require any corresponding reply.
- The radio station is a simplex channel as it transmits the signal to the listeners but never allows them to transmit back.
- Remote control ,TV , Keyboard and Monitor are the examples of the simplex mode as a keyboard can only accept the data from the user and monitor can only be used to display the data on the screen


Transmission in only one direction

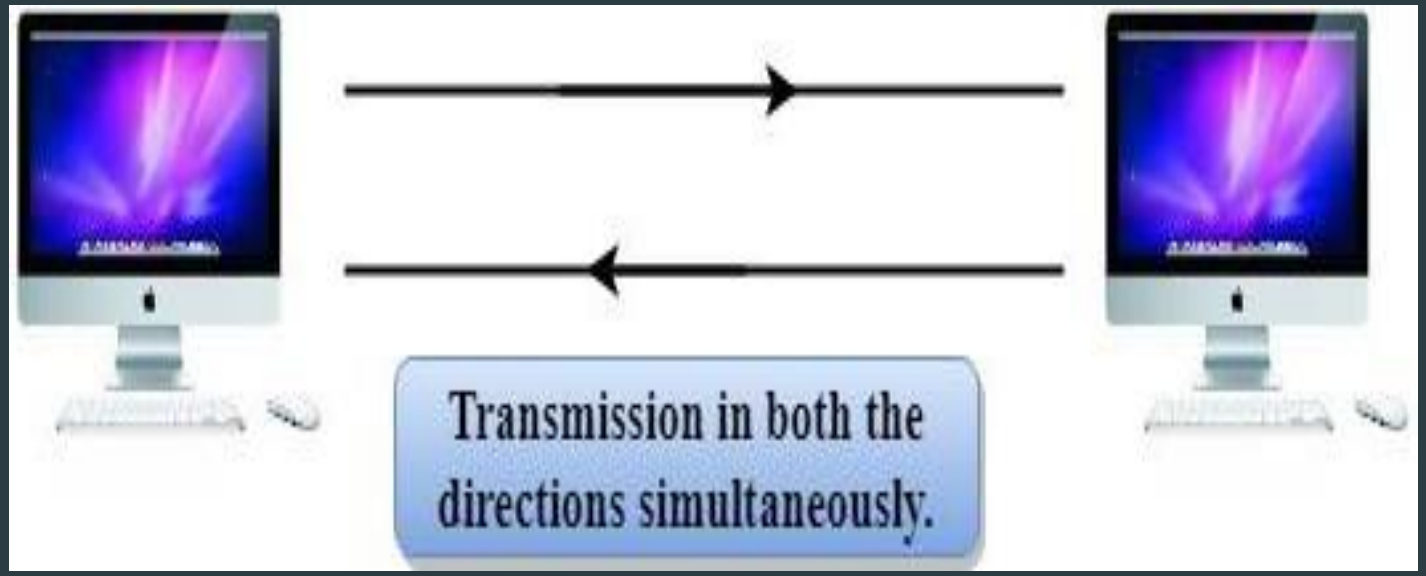
2. Half-Duplex mode:

- In a Half-duplex channel, direction can be reversed, i.e, the station can transmit and receive the data as well.
- Messages flow in both the directions, but not at the same time.
- The entire bandwidth of the communication channel is utilized in one direction at a time.
- In half-duplex mode, it is possible to perform the error detection, and if any error occurs, then the receiver requests the sender to retransmit the data.
- A Walkie-talkie is an example of the Half-duplex mode. In Walkie-talkie, one party speaks, and another party listens. After a pause, the other speaks and first party listens. Speaking simultaneously will create the distorted sound which cannot be understood.


Transmission in either direction, but not simultaneously

3. Full-duplex mode:

- In Full duplex mode, the communication is bi-directional, i.e., the data flow in both the directions.
- Both the stations can send and receive the message simultaneously.
- Full-duplex mode has two simplex channels. One channel has traffic moving in one direction, and
- another channel has traffic flowing in the opposite direction.
- The Full-duplex mode is the fastest mode of communication between devices.
- The most common example of the full-duplex mode is a telephone network. When two people are communicating with each other by a telephone line, both can talk and listen at the same time.


Transmission in both the directions simultaneously.

Introduction of data communication

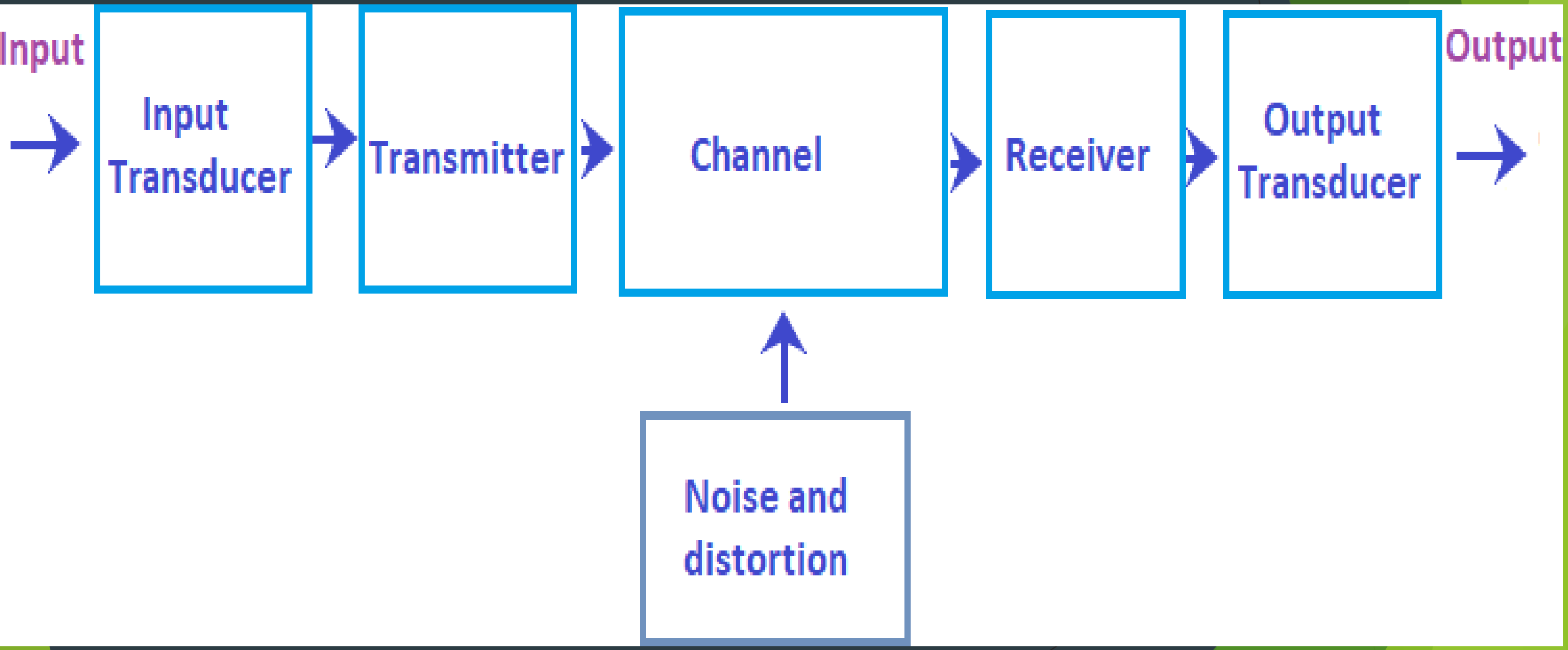
Data communications are the exchange of data between two devices using one or multiple forms of transmission medium. i.e data communication is movement of data from one device or end- point to another device or end point through electrical or optical medium.

- A data communications system has five components:

1. **Message**: The message is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.
2. **Sender**: The sender is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.
3. **Receiver**: The receiver is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.
4. **Transmission medium:** The transmission medium is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves.
5. **Protocol**: A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices.

Analog Communication System:

- The message is transmitted in the form of analog signal.



1. **Input transducer**: The input transducer converts the information in the message signal into the electrical energy suitable for transmission. The sources of information are audio, television, computers, etc.

2. **Transmitter**: The transmitter converts the electrical signal into a form suitable for transmission for the channel. It performs modulation by superimposing the message signal on the high-frequency carrier signal.

3. **Communication channel**: Communication channel is a medium to transmit the electrical signal from transmitter to the receiver.

4. **Noise and Distortion**: It is external Signal Combine with the Message Signal

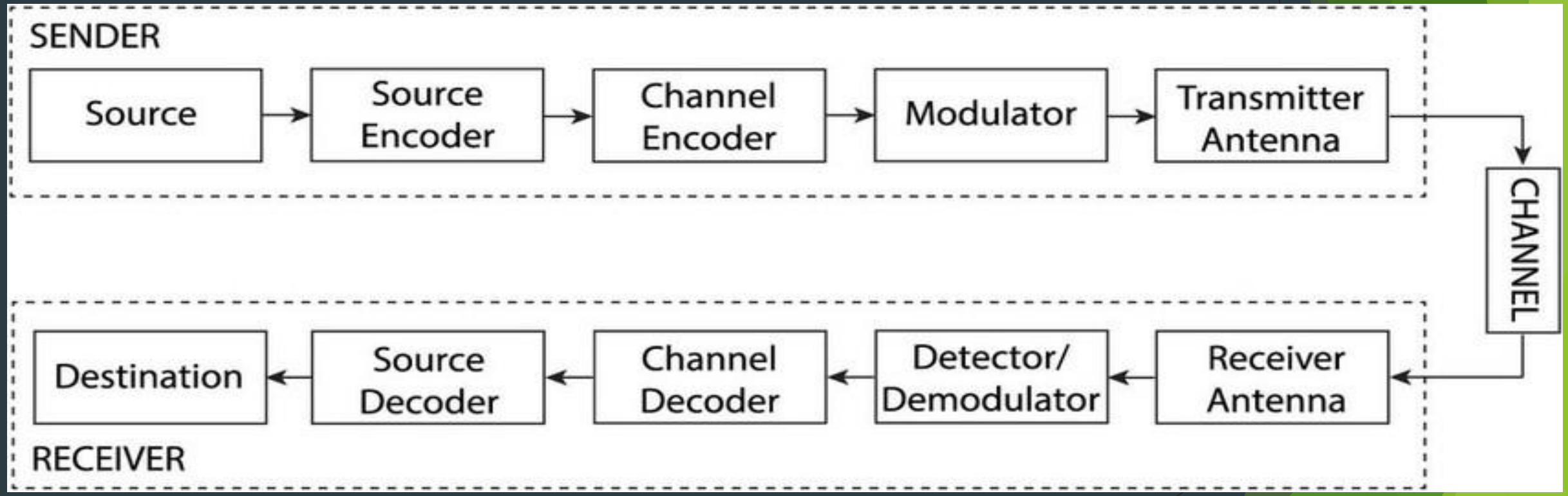
5. **Receiver:** The receiver receives information from the channel. It extracts the necessary information from the signal required by the output transducer. The receiver performs the opposite of modulation and multiplexing, i.e., demodulation.

6. **Output transducer:** The output transducer works reversely as that of the input transducer. It converts the electrical energy into the original signal.

- The loudspeakers convert the electrical energy to sound.
- The motors convert the electrical energy into motion.
- The LEDs (Light Emitting Diodes) convert the electrical energy to light energy.

Digital Communication System

In this the baseband message is to be transmitted in the form of a digital signal.
Basic Blog diagram of Digital communication system:



1. Source: It is where the information to be transmitted is originates. Here the information may be available in the digital form or the analog form.

2. **Source encoder**: The main task of source encoding is to reduce redundancy so we can use bandwidth effectively. This helps in effective utilization of the bandwidth. It removes the redundant bits or unnecessary excess bits that is zeros from the input data.

3. **Channel Encoder** : The channel encoder, does the coding for error correction. During the transmission of the signal, due to the noise in the channel, the signal may get altered. Hence, the channel encoder provides some redundant bits to the message data in order to provide an error free data on the receiver side.

4. **Modulator:** To transmit information at longer distance we need to convert the low frequency digital signal into high frequency analog signal. Digital Modulator multiplies the digital signal we converted in last stage with high frequency carrier signal and convert it in high frequency analog signal so we can easily send it by antenna.

5. **Demodulator**: It will demodulate the signal that received by the receiver antenna. And again convert the high frequency analog signal to digital signal.

6.**Channel Decoder**: The channel decoder, after detecting the sequence, does some error corrections. During the transmission, the signal might get distorted. This is corrected by adding some redundant bits to the signal. This addition of bits helps in the complete recovery of the original signal.

7. **Source Decoder**: Source decoder will convert the digital signal into the analog signal. And that will again converted into information signal by the speaker, television or any other device. It fully decodes the signal and extracts the actual signal.

Advantages of Digital Communication system:

1. It is Simple and cheaper because of integrated circuits became smaller, speedy and cheap.
2. More privacy and security through the use of encryption because, we can re-arrange digital data.
3. Data correction, error detection and error correction is possible.
4. Using multiplexing techniques many data can be merged and transmitted over common channel.
5. Large amount of noise can be tolerated.

Disadvantage of Digital Communication System:

1. High power consumption due to multiple stages and complex circuit as we show in the Digital Communication Block Diagram.
2. Increased Bandwidth
3. Analog and digital Converter requires for the conversion.
4. Synchronization is compulsory, if we don’t synchronize the data so there will be many errors in information

Advantages of analog communication:

The advantages of analog communication are as follows:

1. Analog signal uses less bandwidth as compared to the digital signal. It is due to the use of amplifier in the analog communication system, which improves the signal and reduces the distortion.
2. It provides a more accurate method of representation due to its continuous nature.

Assignment-1

- Differences between Analog and digital communication System
- Standard Organization
- Evolution of data communication System.

Evolution of Data Communication

- The history of data communications actually dates back almost 200 years, back to the years without Wi-Fi or cell phones.
- In 1837, Samuel Morse's invention of the telegraph began the history of data communication.
  - This once remarkable invention sent signals over a series of wires from place to place.
  - The Great Western Railroad adopted the telegraph service in 1843. This allowed the telegraph service to expand across the United States.
  - 1843- Alexander Bain patented a Printing Telegraph, Telegraph service used by the Great western Railway, an endorsement that allowed the service to expand across the nation.
- In 1876, Alexander Graham Bell improved the telegraph with the introduction of the telephone.

Contd….

- It wasn't until 100 years later that telephone lines were able to traffic data. However, Bell's invention laid the groundwork for future data communication inventions.
- In 1958, the U.S. government improved these technological advancements by launching satellites that were communication-oriented.
- Within the following decades, data communications expanded to computer connections.

Cont..

- More than 1 million servers were using Internet Protocol technology by 1991.
- The World Wide Web had also become the primary component of the Internet.

Application of Digital Communication System

- Digital communication systems have numerous applications across a wide range of industries and technologies. Here are some key applications of digital communication systems:

1. **Telecommunications:**

- **Voice and Video Communication**: Digital systems enable high-quality voice calls and video conferencing over networks like VoIP (Voice over IP) and video calling apps.
- **Text Messaging**: SMS and instant messaging apps rely on digital communication for text-based communication.

Continue….

2. **Data Networking:**

- Internet and Wi-Fi
- Local Area Networks (LANs) and Wide Area Networks (WANs)

3. **Broadcasting**:

- Digital Television and Radio
- Streaming Services

4. **Mobile Communications:**

- Cellular Networks :support 2G, 3G, 4G, and 5G cellular networks for mobile voice, data, and multimedia communication.

Continue….

5. **Satellite Communication**:

- Television and Radio Broadcasting
- Global Navigation Satellite Systems (GNSS):GPS system

6. **Secure Communication**:

- Encryption:
- VPNs: to create secure, private channels over public networks.

7. **Education** :Online Learning and E-Learning

THANK YOU !!!!!!